Retraction of the Family Hypothesis from v12: A Self-Correction Note

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Abstract

In v12 §3 we proposed a one-parameter family $z_c(n) = 1/\sqrt{3}+n/(3\pi)$, $w(n) = (n+1)/(3\pi)$ and noted that its $n = 0$ rung matches the v10 lock (DR1 best-fit) and its $n = 1$ rung matches the DR2 free-fit, both within $\sim 0.1\sigma$. In v12 we characterised this match as "a stronger structure" and used it to re-frame the v10 lock failure under DR2 not as a falsification of the program but as the discovery of a discrete two-rung family. v13 and v14 inherited this framing implicitly: both papers refer to the C1 model as "v12 family $n = 1$".

In this v15 note we retract the family hypothesis. The reasons are not data-driven — no new measurement falsifies the family — but methodological. A one-parameter algebraic family fit to two empirical points cannot be statistically meaningful: infinitely many such families pass through any pair of points, and the choice of one over another is not constrained by the data. Presenting the family as a *discovered structure* in v12, rather than as one among many possible algebraic continuations, was a methodological error of the type that the v9-to-v10 corrections were intended to prevent.

This retraction does not affect any numerical result in v13 or v14. The C1 lock $z_c = 1/\sqrt{3} + 1/(3\pi)$, $w = 2/(3\pi)$ is justified independently of the family hypothesis: DR2 free-fitting returns $z_c \approx 0.679$, $w \approx 0.208$, and these values happen to lie within 0.7% and 1.9% respectively of the natural-constant combinations $1/\sqrt{3}+1/(3\pi)$ and $2/(3\pi)$. We adopt the natural-constant values as the lock for parsimony. Whether the agreement is coincidental or carries a meaning we cannot determine here is a question we leave open. All ΔBIC values reported in v13 and v14 (in particular the v13 baseline -11.57, the v14 Task 1 value -18.64, and the v14 Task 2 value -11.38) remain unchanged. What changes is only the interpretive framing.

1 What v12 claimed

For reference, the relevant content of v12 was as follows. The DR1 best-fit values for the free-fit Gaussian modulation were $z_c^{\mathrm{DR1}} \approx 0.583$, $w^{\mathrm{DR1}} \approx 0.106$, which v10 had locked at $1/\sqrt{3}, 1/(3\pi)$ to within 0.7%, 0.1%. The DR2 best-fit values were $z_c^{\mathrm{DR2}} \approx 0.679$, $w^{\mathrm{DR2}} \approx 0.208$, both more than 17% away from the v10 lock and incompatible with it at the $\sim 2.4\sigma$ level on z_c. v12 identified this as a partial failure of the v10 lock under DR2 and reported it openly.

v12 then observed that the DR2 best-fit values match the combinations

$$\left(\frac{1}{\sqrt{3}} + \frac{1}{3\pi}, \frac{2}{3\pi}\right) \approx (0.6834,\ 0.2122) \tag{1}$$

within $\sim 0.1\sigma$, and proposed the family

$$z_c(n) = \frac{1}{\sqrt{3}} + n \cdot \frac{1}{3\pi}, \qquad w(n) = (n+1) \cdot \frac{1}{3\pi}, \qquad n = 0, 1, 2, \ldots \tag{2}$$

so that v10 corresponds to $n = 0$ and the DR2 best-fit to $n = 1$. The v12 abstract called this "a stronger structure" and v12 §3 referred to it as "the v10 family".

v12 §6 (Limitations) acknowledged honestly that the family was post-hoc, that no first-principles derivation existed, and that "different data releases prefer different rungs" was an empirical concern that DR3 would test. However, the family was nevertheless treated as a result of the paper rather than a speculation.

2 Why the family hypothesis fails as a result

2.1 Two points cannot determine an algebraic family

The family was constructed to pass through two empirical points: the DR1 best-fit and the DR2 best-fit. Any one-parameter algebraic family that passes through these two points is consistent with the data. The number of such families that can be written using π, $\sqrt{3}$, e, $\ln 2$ and small integers is large. We did not perform a systematic search; we proposed the first natural-looking algebraic continuation we found and treated it as the structure of nature.

This is a form of post-hoc curve fitting in which the curve is algebraic rather than numerical. Standard statistical reasoning applies: a result that emerges only after the data are seen, and that is not pre-registered, cannot be assigned its nominal significance. The match at $n = 0$ and $n = 1$ to the v10 and DR2 best-fits within 0.1σ is not unlikely under many possible two-point algebraic constructions; it is, in fact, what we *constructed* the family to do.

2.2 The family is not falsifiable in the relevant regime

A genuine prediction would specify the rung that DR3 should select *before* DR3 is observed. v12 did not make such a commitment beyond noting that "$n \in \{0, 1\}$ are favored by current data". After DR3 is observed, three outcomes are possible:

1. DR3 prefers a rung close to $n = 1$. This would be consistent with the family but is also consistent with DR3 being a refinement of DR2 with no fundamentally new information about the underlying modulation.
2. DR3 prefers a different integer rung, e.g. $n = 2$. v12 and its successors could plausibly claim "family confirmed at a new rung," but would be open to the criticism that $n = 2$ was already on the v12 list of rungs and would have been described as a prediction had it been the result.
3. DR3 prefers a non-integer or far-from-family value. The family hypothesis would have to be either abandoned or modified, and any modification (e.g. a different algebraic generator) would be a further post-hoc step.

In all three outcomes the family hypothesis can be rescued post-hoc, which is to say it does not function as a falsifiable proposal. This is precisely the property that v12 §6 noted but did not act on.

2.3 Why v12 should not have presented the family as a result

The v9-to-v10 transition documented an analogous methodological error in the opposite direction: v9 had counted parameters asymmetrically, inflating its BIC margin from -2.15 to -3.08. v10 corrected this openly, downgrading the conclusion. The same standard, applied to v12, would have required the family hypothesis to be presented as a *speculation* rather than a result, with a single sentence or short paragraph in the discussion noting the algebraic match and explicitly declining to treat it as evidence. Instead, v12 promoted the family to a section of its own (§3) and adopted the language of structure and discovery in the abstract. That promotion was the methodological error we now retract.

3 What changes and what does not

3.1 Numerical results: nothing changes

Every numerical result in v13 and v14 is independent of the family hypothesis. The C1 model is fit at fixed values $z_c = 1/\sqrt{3} + 1/(3\pi)$ and $w = 2/(3\pi)$ in all of v13 §3.1 through §3.16 and v14 Task 1 and Task 2. Nothing in those analyses uses the family relation $z_c(n)$, $w(n)$ at $n \neq 1$ or compares C1 to other rungs. The single exception is v13 §3.10, which fits a free n as a continuity check; the result there ($n_{\text{best}} = 0.96$ on DR2) does not depend on the family being a structural claim and remains valid as a numerical observation. Table 1 summarises the principal ΔBIC values, which are unchanged.

Table 1: Principal ΔBIC values from v13 and v14. Each is unchanged by this retraction.

Analysis	ΔBIC (C1 vs ΛCDM)
v13 baseline (DR2, 13-bin SN, $N = 37$)	−11.57
v13 §3.6 fσ8 joint	−11.25
v13 §3.7 r_d-free	−10.02
v13 §3.9 DR1+DR2 merged	−12.65
v13 §3.16 with Planck N_{eff} prior	−6.60
v14 Task 1 (PP unbinned, $N = 1724$)	−18.64
v14 Task 2 (PP unbinned + Planck DP, $N = 1727$)	−11.38

3.2 The C1 lock under the new framing

We restate the justification of the C1 lock without invoking the family.

The DR2 free-fit, performed in v12 with five free parameters (H_0, Ω_m, A, z_c, w), returns

$$z_c^{\text{DR2,free}} = 0.679 \pm 0.043, \qquad w^{\text{DR2,free}} = 0.208 \pm 0.062. \tag{3}$$

We observe that

$$\frac{1}{\sqrt{3}} + \frac{1}{3\pi} \approx 0.6834, \qquad \frac{2}{3\pi} \approx 0.2122, \tag{4}$$

which differ from the free-fit values by 0.7% and 1.9% respectively, both within the 1σ free-fit ranges. We adopt these natural-constant combinations as the lock and refer to the resulting three-parameter (H_0, Ω_m, A) model as C1. The choice is parsimonious in the sense that the lock uses only π, $\sqrt{3}$ and small integers and matches the free best-fit at the percent level.

We do not interpret the lock as derived. We do not claim that the agreement of the DR2 best-fit with these specific natural-constant combinations carries a meaning, nor do we claim it is coincidental. The question of meaning is left open and is to be addressed, if at all, by future data (DR3, full Planck likelihood, perturbation theory) and not by the present paper.

3.3 Reading v13 and v14 under the new framing

v13 and v14 papers should be read with the following terminological substitution: where the text says "C1 (v12 family $n = 1$ rung)" or "the family $n = 1$ lock," read instead "C1 (the DR2 lock)." The numerical content of those papers is unaffected by this substitution. We do not issue corrected versions of v13 or v14; this v15 note serves as the authoritative interpretive context for both.

4 What this retraction is not

4.1 It is not a falsification of C1

The C1 model itself — its functional form, its lock values, its ΔBIC performance — is unaffected. C1 still fits the DR2 baseline data better than ΛCDM by $\Delta\text{BIC} = -11.57$, and the v14 stress tests on Pantheon+ unbinned and Planck distance priors give -18.64 and -11.38 respectively. These are properties of the lock, not of the family.

4.2 It is not a claim that natural-constant locks are inherently illegitimate

The methodological problem is specifically with the family hypothesis, which inferred a structural pattern from two points. A single natural-constant lock that happens to match a single best-fit value is a parsimonious convention, not a structural claim, and is on much firmer ground.

4.3 It is not a return to v10

v10's lock $z_c = 1/\sqrt{3}$, $w = 1/(3\pi)$ was falsified by DR2 and is not revived. C1 stands as the operational lock for all post-DR2 analyses (v13, v14, future work), justified by the DR2 free-fit agreement and not by membership in any family.

5 What we now expect of DR3

A genuine pre-registration of expectations for DR3, made in the spirit avoided by v12, is as follows.

1. We do not predict the DR3 free-fit values of z_c and w. DR3 will provide its own posterior on these parameters, and that posterior is the relevant quantity, not whether DR3 happens to prefer a particular family rung.
2. If DR3's free-fit on (z_c, w) remains compatible with the C1 lock (within, say, 1σ on each), the lock survives.
3. If DR3's free-fit is statistically incompatible with the C1 lock at the $\geq 2\sigma$ level on at least one of (z_c, w), the C1 lock is falsified by DR3 in the same way the v10 lock was falsified by DR2. In that case we will report the new free-fit and will not propose a new natural-constant match without an independent, pre-registered justification.
4. Repetition of the v12 pattern — finding a new natural-constant combination that matches the DR3 best-fit and proposing it as a new "rung" — is a possibility we explicitly disavow. If the C1 lock fails under DR3, the program either finds a first-principles derivation that motivates a new lock or it accepts that no locked-constant version of the modulation is supported by the data.

6 Closing

The strongest single feature of this program through v9 and v10 was the willingness to issue self-corrections (the v9 BIC accounting correction in v10, the disclosure of v10 lock failure in v12). We believe these corrections, more than any of the BIC margins, have been the basis for whatever credibility the program has accumulated. The family hypothesis has been an exception: it was a structural claim made on inadequate evidence and carried forward through v13 and v14. We retract it now, while DR3 has not yet returned, so that the retraction is methodological rather than reactive.

The C1 lock, the v13 stress-test survey, and the v14 Pantheon+ and Planck-distance-prior analyses stand. The interpretive claim that the DR1 and DR2 best-fits constitute two rungs of a discrete family does not.